

 LEVEL-II (C.W)

FORMATION OF DIFFERENTIAL EQUATIONS

- A curve passes through the point (4,2) and at any point (x,y) on it the product of its slope and the ordinate is equal to abscissa of the curve is
1. parabola 2. Ellipse 3. Circle 4. Hyperbola
- The differential equation of the family of circles whose centre lies on x-axis and passing through origin is
1. $x^2 + y^2 + \frac{dy}{dx} = 0$
2. $(y^2 - x^2)dx - 2xydy = 0$
3. $y^2dx + (x^2 + 2xy)dy = 0$
4. $xdy + ydx + x^2dx + y^2dy = 0$
- The D. E of the family of parabolas of which has a latusrectum 4a and whose axes are parallel to x-axis
1) $y_1^3 + 2ay_2 = 0$ 2) $y_1^3 + ay_2 = 0$
3) $y_1^3 + 4ay_2 = 0$ 4) $y_1^3 + 3ay_2 = 0$
- Assume that a spherical rain drop evaporates at a rate proportional to its surface area, the differential equation involving the rate of change of the radius of rain drop
1) $\frac{dr}{dt} = k$ 2) $\frac{dr}{dt} = -k$
3) $\frac{d^2r}{dt^2} = k$ 4) $\frac{d^2r}{dt^2} = -k$
- The D.E whose solution is $y = c_1e^x + c_2e^{-x} + c_3 \sin x + c_4 \cos x$ is
1. $y_4 = y_2$ 2. $y_4 = y_3$ 3. $y_4 = y_1$ 4. $y_4 = y$
- The D.E whose solution is $y = c^2 + \frac{c}{x}$ is
1. $y = x^4y_1 - xy_1^2$ 2. $y = x^4y_1^2 + xy_1$
3. $y = x^4y_1^2 - xy_1$ 4. $y = x^4y_1^2 + 2xy_1$

DIFFERENTIAL EQUATIONS

- The differential equation of all non-vertical lines in a plane is
1) $\frac{d^2y}{dx^2} = 0$ 2) $\frac{d^2x}{dy^2} = 0$ 3) $\frac{dy}{dx} = 0$ 4) $\frac{dx}{dy} = 0$
- The D.E whose solution is $y = a \cos x + b \sin x + x \sin x$ is
1. $y_2 + y = \cos x$ 2. $y_2 + y = \sin x$
3. $y_2 + y = 2 \sin x$ 4. $y_2 + y = 2 \cos x$
- The D.E whose solution is $xy = ae^x + be^{-x}$ is
1. $xy_2 + 2y_1 = xy$ 2. $xy_2 - 2y_1 = xy$
3. $xy_2 - 2y_1 + xy = 0$ 4. $xy_2 + 2y_1 + xy = 0$
- Differential equation having general solution $y = (\sin^{-1}x)^2 + A(\cos^{-1}x) + B$ where A, B are arbitrary constants is
1) $(1-x^2)y_2 - xy_1 = 0$ 2) $(1-x^2)y_2 - xy_1 = 1$
3) $(1-x^2)y_2 - xy_1 = 2$ 4) $(1-x^2)y_2 - xy_1 = 4$
- A normal at any point (x,y) to the curve $y = f(x)$ cuts a triangle of unit area with the axis, the differential equation of the curve is
1. $y^2 - x^2 \left(\frac{dy}{dx} \right)^2 = 4 \frac{dy}{dx}$
2. $x^2 - y^2 \left(\frac{dy}{dx} \right)^2 = \frac{dy}{dx}$
3. $y^2 \left(\frac{dy}{dx} \right)^2 + 2(xy-1) \frac{dy}{dx} + x^2 = 0$
4. $x + y \left(\frac{dy}{dx} \right) = y$

SOLUTIONS OF THE DIFFERENTIAL EQUATIONS VARIABLE SEPARABLE

- If $\frac{dy}{dx} = e^{-2y}$ and $y=0$ when $x=5$, the value of x for $y=3$ is
1) e^5 2) $e^6 + 1$ 3) $\frac{e^6 + 9}{2}$ 4) $\log_e 6$

13. The differential equation

$y \frac{dy}{dx} + x = A$ where A is a constant represents a set of

- 1) circles centre at y -axis
- 2) circles centre at x -axis
- 3) parabolas
- 4) ellipses

14. The family passing through (0, 0) and satisfying

the differential equation $\frac{y_2}{y_1} = 1$ (where

$y_n = \frac{d^n y}{dx^n}$) is

- 1) $y = k$
- 2) $y = kx$
- 3) $y = k(e^x + 1)$
- 4) $y = k(e^x - 1)$

15. The solution of $\frac{dy}{dx} = x \log x$

- 1) $2y = x^2 \left[\log x + \frac{1}{2} \right] + c$
- 2) $2y = x^2 \left[\log x - \frac{1}{2} \right] + c$
- 3) $y = \frac{x^2}{2} (\log 2 - x) + c$
- 4) $y^2 = x^2 \log x + x + c$

16. The Solution of $\sqrt{1-x^2} \sin^{-1} x dy + y dx = 0$

- 1) $y \tan^{-1} x = c$
- 2) $y \sin^{-1} x = c$
- 3) $y \cos^{-1} x = c$
- 4) $x \sin^{-1} y = c$

17. The solution of

$x dx + y dy + (x^2 + y^2) dy = 0$

1. $(x^2 + y^2) e^{2y} = c$
2. $(x^2 + y^2) = cxy$
3. $(x^2 + y^2) = cx^2$
4. $(x^2 + y^2) e^{2x} = c$

18. The solution of $x dx + y dy = x^2 y dy - xy^2 dx$ is

1. $x^2 - 1 = c(1 + y^2)$
2. $x^2 + 1 = c(1 - y^2)$
3. $x^3 - 1 = c(1 + y^3)$
4. $x^3 + 1 = c(1 - y^3)$

19. The solution of $(x + y + 1) \frac{dy}{dx} = 1$ is

1. $x = -(y + 2) + ce^y$
2. $y = -(x + 2) + ce^x$
3. $x = -(y + 2) + ce^x$
4. $x = (y + 2) + ce^{-y}$

20. Solution of $x + y = \cos^{-1} \left(\frac{dy}{dx} \right)$ is

- 1) $x + c = \tan \left(\frac{x + y}{2} \right)$
- 2) $x + c = \sin \left(\frac{x + y}{2} \right)$
- 3) $x + c = \sec \left(\frac{x + y}{2} \right)$
- 4) $x + c = \cos ec \left(\frac{x + y}{2} \right)$

21. Solution of $(x - y)^2 \frac{dy}{dx} = a^2$

- 1) $2y = c + a \log \left(\frac{x - y - a}{x - y + a} \right)$
- 2) $y = c + a \log \left(\frac{x - y + a}{x - y - a} \right)$
- 3) $2y = c - a \log \left(\frac{x - y}{x + y} \right)$
- 4) $2y^2 = c + \log \left(\frac{x - y + a}{x - y - a} \right)$

22. The solution of $\frac{dy}{dx} + 1 = e^{x+y}$ is

- 1) $e^{-(x+y)} + x + c = 0$
- 2) $e^{-(x+y)} - x + c = 0$
- 3) $e^{x+y} + x + c = 0$
- 4) $e^{x+y} - x + c = 0$

23. Equation of the curve whose gradient at any

point (x, y) on it is $\frac{x-a}{y-b}$ and which passes through the origin is

- 1) $x^2 - y^2 = 2(ax - by)$
- 2) $x^2 + y^2 = 2(ax + by)$
- 3) $x^2 - y^2 = 2(bx + ay)$
- 4) $x^2 + y^2 = 2(ax - by)$

24. Equation of the curve whose sub tangent is constant is

- 1) $y^2 = ce^{\frac{x^2}{k}}$
- 2) $y = cx^2$
- 3) $y = ce^{\frac{x}{k}}$
- 4) $y = ce^{x^2}$

25. The solution of

$$(1 + \sin^2 x) dy + \cos x (1 + y^2) dx = 0 \text{ given that}$$

$$y = 2 \text{ when, } x = \frac{\pi}{2}$$

$$1) y = \frac{\sin x + 3}{3 \sin x - 1} \quad 2) y = \frac{3(\sin x - 1)}{\sin x + 3}$$

$$3) y = \frac{3 \sin x + 1}{\sin x - 3} \quad 4) y = \frac{\sin x + 3}{\sin x + 1}$$

26. Equation of the curve passing through $(0, 0)$

$$\text{and satisfying the equation } \frac{dy}{dx} = (x - y)^2$$

$$1) e^{2x} (1 - x + y) = 1 + x - y$$

$$2) e^{2x} (1 + x - y) = 1 - x + y$$

$$3) e^{2x} (1 - x + y) = -(1 + x + y)$$

$$4) e^{2x} (1 + x + y) = 1 - x + y$$

27. The solution of $e^x \sqrt{1 - y^2} dx + \frac{y}{x} dy = 0$

$$1) (x - 1)^2 e^x = (1 - y^2) + c$$

$$2) (x + 1)e^x = \sqrt{1 - y^2} + c$$

$$3) x.e^x = \sqrt{1 - y^2} + c$$

$$4) (x - 1)e^x = \sqrt{1 - y^2} + c$$

28. The solution of $\sin^{-1} y dx + \frac{x}{\sqrt{1 - y^2}} dy = 0$ is

$$1) y \sin^{-1} x = c \quad 2) y = c \sin^{-1} x$$

$$3) y = \sin\left(\frac{c}{x}\right) \quad 4) x = c \sin y$$

HOMOGENEOUS EQUATIONS

29. The solution of

$$(x^3 - 3xy^2) dx = (y^3 - 3x^2y) dy$$

$$1) y^2 - x^2 = c(x^2 + y^2)^2 \quad 2) y^3 - x^3 = c(x^2 + y^2)$$

$$3) y^2 + x^2 = c(x^2 - y^2) \quad 4) y^3 + x^3 = c(x^2 - y^2)$$

NON-HOMOGENEOUS EQUATIONS

30. The solution of

$$(2x + 3y - 5) dx + (3x - 4y + 1) dy = 0$$

$$1) x^2 + 3xy - 2y^2 - 5x + y = c$$

$$2) x^2 + 3xy - 4y^2 - 5x - y = c$$

$$3) x^2 + 3y^2 - 2xy + 5x + y = c$$

$$4) x^2 - 3y^2 - 2xy + 5x + y = c$$

31. The solution of $\frac{dy}{dx} = \frac{x - 2y + 3}{2x - y + 5}$

$$1) x^2 - 2xy + y^2 + 3x - 5y = c$$

$$2) x^2 - 4xy + y^2 + 6x - 10y = c$$

$$3) x^2 - 4xy - y^2 - 3x + 5y = c$$

$$4) x^2 - 2xy + 2y^2 + 6x - 5y = c$$

32. The solution of

$$(6x + 7y - 4) dx + (7x - 4y + 3) dy = 0$$

$$1) 3x^2 + 7xy - 2y^2 - 4x + 3y = c$$

$$2) 6x^2 - 4y^2 + 7xy - 4x + 3y = c$$

$$3) 3x^2 + 14xy - 4y^2 - 4x + 3y = c$$

$$4) 6x^2 + 4x - 4y^2 + 4x + 3y = c$$

INTEGRATING FACTORS OF DIFFERENTIAL EQUATION

33. I.F of $x \frac{dy}{dx} + y(1 + x) = 1$

$$1) x.e^x \quad 2) \frac{e^x}{x} \quad 3) x + \log x \quad 4) x \log x$$

34. I.F of $x \sin x \frac{dy}{dx} + y(x \cos x + \sin x) = \sin x$

$$1) x \cos x \quad 2) x \sec x \quad 3) x \sin x \quad 4) x \cos ecx$$

LINEAR DIFFERENTIAL EQUATIONS

35. The general solution of the differential

$$\text{equation } y dx - (x + 2y^2) dy = 0$$

$$1. y = 2x^2 + cx \quad 2. x = 2y^2 + cy$$

$$3. x^2 = y^2 + cy \quad 4. y = x^2 + c$$

36. $y + x^2 = \frac{dy}{dx}$ has the solution

1. $y + x^2 + 2x + 2 = c.e^x$
2. $y + x + 2x^2 + 2 = c.e^x$
3. $y + x + x^2 + 2 = c.e^{2x}$
4. $y^2 + x + x^2 + 2 = c.e^{2x}$

37. If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and

$y(0) = -1$ then $y(1)$ is equal to

- 1) $-\frac{1}{2}$
- 2) $e + \frac{1}{2}$
- 3) $e - \frac{1}{2}$
- 4) $\frac{1}{2}$

38. $x \frac{dy}{dx} + y - x + xy \cot x = 0$ ($x \neq 0$)

1. $y(x \sin x) = -x \cos x + \sin x + c$
2. $x(x \sin x) = x \cos x - \sin x + c$
3. $y \sin x = -x \cos x + 2 \sin x + c$
4. $y \cos x = x \sin x + 2 \cos x + c$

39. The solution of $\sin 2x \left(\frac{dy}{dx} \right) - y = \tan x$

- 1) $x - y \sin x = c$
- 2) $xy \tan x = c$
- 3) $y = \tan x + c$
- 4) $y = \tan x + c\sqrt{\tan x}$

BERNOULLI EQUATION

40. Which of the following transformation reduces the differential equation

$$\frac{dz}{dx} + \frac{z}{x} \log z = \frac{z}{x^2} (\log z)^2 \text{ into the form}$$

$$\frac{du}{dx} + uP(x) = Q(x)$$

- 1) $u = \log z$
- 2) $u = e^z$
- 3) $u = (\log z)^{-1}$
- 4) $u = (\log z)^2$

41. The solution of $\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$ is

given by $x^2 - 2 \tan y + ce^{-x^2} = \lambda$ where λ

- 1) -2
- 2) -1
- 3) 1
- 4) 2

LEVEL-II (C.W.) - KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) 4 | 2) 2 | 3) 1 | 4) 2 | 5) 4 | 6) 3 | 7) 1 |
| 8) 4 | 9) 1 | 10) 3 | 11) 3 | 12) 3 | 13) 2 | 14) 4 |
| 15) 2 | 16) 2 | 17) 1 | 18) 1 | 19) 1 | 20) 1 | 21) 1 |
| 22) 1 | 23) 1 | 24) 3 | 25) 1 | 26) 1 | 27) 4 | 28) 3 |
| 29) 1 | 30) 1 | 31) 2 | 32) 1 | 33) 1 | 34) 3 | 35) 2 |
| 36) 1 | 37) 1 | 38) 1 | 39) 4 | 40) 3 | 41) 3 | |

LEVEL-II (C.W.) - HINTS

1. $y.m = x \Rightarrow y \cdot \frac{dy}{dx} = x ; xdx - ydy = 0$
2. Required circle is $x^2 + y^2 + 2gx = 0$
3. $(y - \beta)^2 = 4a(x - \alpha)$ eliminate parameters α, β
4. $\frac{dv}{dt} \alpha s \Rightarrow -\frac{d}{dt} \left(\frac{4}{3} \pi r^3 \right) = 4\pi r^2$
5. Find y_4
6. Find y_1
7. $y = mx + c \therefore \frac{d^2 y}{dx^2} = 0$
8. Find y_2
9. differentiate two times with respect to x
10. $y_1 = \frac{2\sin^{-1}x}{\sqrt{1-x^2}} - \frac{A}{\sqrt{1-x^2}}$
 $\Rightarrow \sqrt{(1-x^2)} y_1 = 2\sin^{-1}x - A$
 again differentiate $(1-x^2) y_2 - xy_1 = 2$
11. Equation of normal at point p is
 $Y - y = \frac{-dx}{dy} (X - x)$ Area of triangle OAB = 1
 $\frac{1}{2} \left(y \frac{dy}{dx} + x \right) \left(x \frac{dx}{dy} + y \right) = 1$
 $\left(y \frac{dy}{dx} + x \right) \left(y \frac{dy}{dx} + x \right) = 2 \frac{dy}{dx}$
 $y^2 \left(\frac{dy}{dx} \right)^2 + 2(xy - 1) \frac{dy}{dx} + x^2 = 0$
12. $e^{2y} = 2x + 2c$ and $c = -\frac{9}{2}$
13. $ydy + xdx = Adx$, on integrating both sides
14. $y_2 = y_1 \Rightarrow y_1 = y + c$, integrating on both sides

$$15. \int dy = \int x \log x dx$$

$$16. \int \frac{dy}{y} + \int \frac{dx}{\sqrt{1-x^2} \sin^{-1} x} = c$$

$$17. \int \frac{2xdx + 2ydy}{x^2 + y^2} + \int dy = c$$

$$18. \int \frac{2x}{x^2-1} dx = \int \frac{2y}{y^2+1} dy$$

$$19. \text{ Put } x+y+I = u$$

$$20. \text{ put } x+y = V, \cos V = V + x \frac{dy}{dx}$$

$$21. \text{ put } x-y = V$$

$$22. \text{ put } x+y = u$$

$$23. \frac{dy}{dx} = \frac{x-a}{y-b}$$

$$24. \frac{y}{\frac{dy}{dx}} = c \text{ (lengths of sub tangent) eliminate } C$$

$$25. \frac{dy}{1+y^2} + \frac{\cos x dx}{1+\sin^2 x} = 0$$

$$26. \text{ put } x-y = V$$

$$27. \text{ apply variable separable method}$$

$$28. \frac{dx}{x} + \frac{dy}{\sin^{-1} y \sqrt{1-y^2}} = 0$$

$$29. \text{ put } y = Vx$$

$$30. \frac{dy}{dx} = \frac{-(2x+3y-5)}{3x-4y+1} \text{ convert in the form of homogeneous equation then solve.}$$

$$31. b_1 = -a_2; \text{ taking cross multiplication then apply integration}$$

$$32. 6 \int x dx - 4 \int y dy + 7 \int d(xy) - 4 \int dx + 3 \int dy = c$$

$$33. I.F = e^{\int \frac{1+x}{x} dx}$$

$$34. I.F = e^{\int \frac{x \cos x + \sin x}{x \sin x} dx}$$

$$35. \text{ The G.E } y \frac{dx}{dy} = x + 2y^2 \Rightarrow \frac{dx}{dy} + x \left(-\frac{1}{y} \right) = 2y$$

$$I.F = e^{\int -\frac{1}{y} dy} = e^{-\log|y|} = \frac{1}{y}$$

$$G.S = x \frac{1}{y} = \int (2y) \cdot \frac{1}{y} dy$$

$$\therefore \frac{x}{y} = 2y + c \Rightarrow x = 2y^2 + cy$$

$$36. I.F = e^{\int -dx}, \text{ solution is}$$

$$y(I.F) = \int (I.F) x^2 dx + c$$

$$37. \frac{dy}{dt} - \frac{t}{1+t} y = \frac{1}{1+t}; I.F = e^{\int \frac{-t}{1+t} dt} = (1+t)e^{-t}$$

$$G.S y(1+t)e^{-t} = -e^{-t} + c$$

$$\Rightarrow y(1+t) = -1 + ce^t$$

$$y(0) = -1 \Rightarrow c = 0 \Rightarrow y = -\frac{1}{1+t}; y(1) = -\frac{1}{2}$$

$$38. x \frac{dy}{dx} - x + y(x \cot x + 1) = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x}(x \cot x + 1) = 1$$

$$\frac{dy}{dx} + y \left(\cot x + \frac{1}{x} \right) = 1$$

$$I.F =$$

$$e^{\int \left(\cot x + \frac{1}{x} \right) dx} = e^{\log(\sin x) + \log x} = e^{\log(x \sin x)} = x \sin x$$

$$G.S = y(x \sin x) = -x \cos x + \sin x + c$$

$$39. \text{ dividing with } \sin 2x$$

$$40. \text{ dividing the given equation by } z(\log z)^2$$

$$\text{we have } \frac{1}{z(\log z)^2} \frac{dz}{dx} + \frac{1}{\log z} \cdot \frac{1}{x} = \frac{1}{x^2} \text{ put } \frac{1}{\log z} = u$$

$$41. \text{ Put } \tan y = t, \text{ then } \sec^2 y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\text{Given D.E is } \frac{dt}{dx} + 2xt = x^3, \text{ this a linear equation in } t.$$

$$I.F = e^{\int 2x dx} = e^{x^2}, \text{ solution is } t e^{x^2} = \int e^{x^2} x^3 dx$$

$$\Rightarrow (\tan y) e^{x^2} = \frac{1}{2} e^{x^2} (x^2 - 1) + c$$

$$\Rightarrow 2 \tan y - x^2 + 1 = c e^{-x^2}$$

LEVEL-II (H.W)

FORMATION OF DIFFERENTIAL EQUATIONS

1. The D.E of the family of all circles in the first quadrant touching the coordinate axes.

$$1) (x-y)^2(1+y_1^2) = (x+yy_1)^2$$

$$2) (x+y)^2(1+y_1^2) = (x+yy_1)^2$$

$$3) (x+y)^2(1-y_1^2) = (x+yy_1)^2$$

$$4) (x-y)^2(1+y_1^2) = (x-yy_1)^2$$

2. D.E of the parabolas having its axis has x-axis and origin as focus is

$$y\left(\frac{dy}{dx}\right)^2 + 4x\frac{dy}{dx} = 4 \quad 2. \quad 2x\frac{dy}{dx} - y = 0$$

$$3. \quad y\left(\frac{dy}{dx}\right)^2 + y = 2xy\frac{dy}{dx}$$

$$4. \quad y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} - y = 0$$

3. The D.E of the family of ellipses with centre at the origin and having co-ordinate axes as axes is

$$1) x[yy_2 + y_1^2] = yy_1 \quad 2) x^2y_2 + y_1^2 = y$$

$$3) xy_2 + y_1^2 = y \quad 4) xy_2 + y_1^2 + y = 0$$

4. The D.E whose solution is $y = ax^3 + bx^2$ is

$$1. \quad x^2y_2 - 4xy_1 + 6y = 0 \quad 2. \quad x^2y_2 - 4y_1 + 6y = 0$$

$$3. \quad x^2y_2 + 4xy_1 - 6y = 0$$

$$4. \quad x^2y_2 - 2xy_1 + 6y = 0$$

5. The differential equation of family of curves whose tangent form an angle of $\pi/4$ with the hyperbola $xy = c^2$ is

$$1) \frac{dy}{dx} = \frac{x^2 + c^2}{x^2 - c^2} \quad 2) \frac{dy}{dx} = \frac{x^2 - c^2}{x^2 + c^2}$$

$$3) \frac{dy}{dx} = \frac{-c^2}{x^2} \quad 4) \frac{dy}{dx} = \frac{c^2}{x^2}$$

6. The D.E whose solution is $y = c(x-c)^2$ is

$$1. \quad \left(\frac{dy}{dx}\right)^3 = 4y\left(x\frac{dy}{dx} - 2y\right)$$

$$2. \quad y_1^3 = 2y(xy_1 - y)$$

$$3. \quad \left(\frac{dy}{dx}\right)^3 = 4y\left(2x\frac{dy}{dx} - y\right)$$

$$4. \quad \left(\frac{dy}{dx}\right)^3 = 2y\left(x\frac{dy}{dx} - 4y\right)$$

7. The D.E whose solution is $x^3 = c(y+c)^2$ is

$$1. \quad 8xy_1^3 - 4yy_1^2 = 27x \quad 2. \quad 8xy_1^3 = 12yy_1^2 + 27x$$

$$3. \quad 8xy_1^2 - 12yy_1^3 = 9y \quad 4. \quad 8xy_1^3 = 3(yy_1^2 + 4x)$$

8. The D.E whose solution is $xy = ax^2 + \frac{b}{x}$ is

$$1. \quad x^2y_2 + 2xy_1 = 2y \quad 2. \quad x^2y_2 - xy_1 + 2y = 0$$

$$3. \quad x^2y_2 + xy_1 + y = 0 \quad 4. \quad x^2y_2 + xy_1 + 2y = 0$$

9. The D.E whose solution is

$$y = (a+bx)\cos x + (c+dx)\sin x \text{ is}$$

$$1. \quad y_4 = y \quad 2. \quad y_4 + 2y_2 + y = 0$$

$$3. \quad y_4 = 2y_2 \quad 4. \quad y_4 - 2y_2 + 4y = 0$$

10. The D.E whose solution is $y = ax^2 + bx$ is

$$1. \quad x^2y_2 + 2xy_1 + 2y = 0$$

$$2. \quad x^2y_2 - 2xy_1 + 2y = 0$$

$$3. \quad x^2y_2 - 2xy_1 - 2y = 0$$

$$4. \quad x^2y_2 + 2xy_1 - 2y = 0$$

SOLUTIONS OF THE DIFFERENTIAL EQUATIONS VARIABLE SEPARABLE

11. Equation of the curve passing through the point (3,9) and which satisfies $y_1 = x + \frac{1}{x^2}$

$$1. \quad 6xy = 3x^3 - 6x + 29 \quad 2. \quad 6xy = 3x^3 - 29x + 6$$

$$3. \quad 6xy = 3x^3 + 29x - 6 \quad 4. \quad 6xy = 3x^3 - 29x - 6$$

12. Solution of $\frac{d^2y}{dx^2} = \log x$ is

- 1) $y = \frac{1}{2}x^2 \log x - \frac{3}{4}x^2 + C_1x + C_2$
- 2) $y = \frac{1}{2}x^2 \log x + \frac{3}{4}x^2 + C_1x + C_2$
- 3) $y = \frac{-1}{2}x^2 \log x - \frac{3}{4}x^2 + C_1x + C_2$
- 4) $y = 3x^2 + 4x + c_1$

13. The solution of

$$x\sqrt{1+x^2}dx + y\sqrt{1+y^2}dy = 0$$

- 1) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$
- 2) $(\sqrt{1+x^2}) \times (\sqrt{1+y^2}) = c$
- 3) $(1+x^2)^{\frac{3}{2}} + (1+y^2)^{\frac{3}{2}} = c$
- 4) $\frac{x}{\sqrt{1+x^2}} + \frac{y}{\sqrt{1+y^2}} = c$

14. The solution of $ydx - xdy = xy dx$

- 1) $y = cx^2$
- 2) $y = cx^3$
- 3) $x = cy e^x$
- 4) $y = cx e^x$

15. The solution of $(x^2 + y)dx + (x + y^2)dy = 0$

- 1) $(2x+1) + (2y+1) = c$
- 2) $x^2 + xy + y^2 = c$
- 3) $x^3 + 3xy + y^3 = c$
- 4) $x^3 + xy + y^3 = c$

16. The solution of $\log \left[\frac{dy}{dx} \right] = ax + by$

- 1) $b e^{ax} + a e^{-by} = k$
- 2) $e^{ax} + e^{-by} = c$
- 3) $e^{ax+by} = c$
- 4) $(ax+by) = cxy$

17. Solution of $\frac{dy}{dx} - x \tan(y-x) = 1$

- 1) $\tan(y-x) = c e^{\frac{x^2}{2}}$
- 2) $\sin(y-x) = e^{\frac{x^2}{2}} + c$
- 3) $\sec 2(y-x) c e^{\frac{x^2}{2}}$
- 4) $\cos(y-x) e^{\frac{x^2}{2}} = c$

18. The solution of $\frac{dy}{dx} = e^{3x+y}$ given $y=0$ when $x=0$

1. $e^{3x} + 3e^{-y} = 4$
2. $e^{-y} = e^{3x} + 4$
3. $3e^{-y} = e^{3x} + 12$
4. $y = \frac{e^{3x+y}}{3}$

19. The solution of $x \cos^2 y dx = y \cos^2 x dy$

- 1) $\tan x \tan y = c$
- 2) $y \tan y = x \tan x + c$
- 3) $\tan x \cdot \cos y = \tan y \cos x + c$
- 4) $y \tan y - x \tan x + \log \left(\frac{\cos y}{\cos x} \right) = c$

20. Solution of $\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$

- 1) $\log \left(\frac{x}{1+\sqrt{1+x^2}} \right) + \sqrt{1+x^2} + \sqrt{1+y^2} = c$
- 2) $\log \left(\frac{x}{\sqrt{1+x^2}} \right) + \sqrt{1-x^2} + \sqrt{1+y^2} = c$
- 3) $\log \left(\frac{x}{\sqrt{1+x^2}} \right) = c$
- 4) $\log = (\sqrt{1+x^2} - \sqrt{1+y^2}) + \log \left(\frac{x}{\sqrt{1+x^2}} \right) = c$

21. The solution of $\frac{dy}{dx} = xy + 2x - 3y - 6$

- 1) $(y+2)^2 = c.e^{(x-3)^2}$
- 2) $\log(y+2) = x^2 - 3x + c$
- 3) $y+2 = 2(x-3) + c$
- 4) $(y+2)(x-3) = c$

22. Solution of $y - x \frac{dy}{dx} = 3 \left[1 - x^2 \frac{dy}{dx} \right]$

- 1) $(y+3)(1+3x) = cx$
- 2) $(y-3)(1-3x) = cx$
- 3) $(y-3)(1+3x) = cx$
- 4) $(y+3)(1-3x) = cx$

23. Find the equation of the curve whose D.E is

$(1+y^2)dx = xydy$ and passing through $(1,0)$ is

- 1) $x^2 - y^2 = 1$
- 2) $4x^2 - y^2 = 4$
- 3) $x^2 + y^2 = 1$
- 4) $4x^2 + y^2 = 4$

24. Solution of $x e^{x^2+y} . dx = y . dy$

- 1) $x.e^{x^2} + 2y.e^y = c$ 2) $e^{x^2} + 2(y+1)e^{-y} = c$
3) $x.e^{x^2} + y = 2y + c$ 4) $x.e^{x^2} y = c$

HOMOGENEOUS EQUATIONS

25. On putting $y = vx$ the equation

$x^2 dy + y(x+y) dx = 0$ transformed to

- 1) $x dv + (v^2 + 2v) dx = 0$
2) $v dx + (2x + x^2) dv = 0$
3) $v^2 dx = (x + x^2) dv$
4) $v dv + (2x + x^2) dx = 0$

26. The solution of

$$\left[x + y \sin \left(\frac{y}{x} \right) \right] dx = x \sin \left(\frac{y}{x} \right) dy$$

- 1) $\cos \left(\frac{y}{x} \right) + \log |x| = c$ 2) $\sin \left(\frac{y}{x} \right) + \frac{1}{x} = c$
3) $x \cos \left(\frac{y}{x} \right) + \log |x| = c$
4) $x \sin \left(\frac{y}{x} \right) + \log |x| = c$

LINEAR DIFFERENTIAL EQUATIONS

27. I.F of $x \frac{dy}{dx} = (2y + 2x^4 + x^2)$

- 1) x^{-2} 2) x^{-1} 3) x 4) x^2

28. I.F of $(x + 2y^3) \frac{dy}{dx} = y^2$

1. $e^{1/y}$ 2. $e^{-1/y}$ 3. $3y$ 4. $-\frac{1}{y}$

29. I.F of $1 + (x \tan y - \sec y) \frac{dy}{dx} = 0$

- 1) $\tan y$ 2) $\sec y$ 3) $\tan x$ 4) $\sec x$

30. A particular solution of the differential equation $(x+1) \frac{dy}{dx} = 2e^{-y} + 1$ **given that** $y = 0$ **and** $x = 0$

1. $\frac{x}{2+e^y} = -\frac{1}{2+e^y} + c$ 2. $\frac{x}{2+e^{-y}} = \frac{1}{2+e^y} + c$
3. $\frac{x}{2+e^y} = \frac{1}{2+e^x} + c$ 4. $\frac{y}{2+e^x} = x + c$

31. The solution of

$$\frac{dy}{dx} + ay = e^{mx} \text{ is (where } a + m = 0)$$

- 1) $e^{ax} y = x + c$ 2) $e^{ax} y = y + c$
3) $e^{ay} x = y + c$ 4) $e^{ay} y = x + c$

32. $3x^2 y + x^3 \frac{dy}{dx} = 3 \cos x \Rightarrow x^3 y =$

1. $\sin x$ 2. $2 \sin x + c$
3. $3 \sin x + c$ 4. $3 \cos x + c$

33. The solution of the differential equation

$$\frac{dy}{dx} - y \tan x = e^x \sec x \text{ is}$$

- 1) $y = e^x \cos x + c$ 2) $y \cos x = e^x + c$
3) $y = e^x \sin x + c$ 4) $y \sin x = e^x + c$

34. $\frac{dy}{dx} + 2y \tan x = \sin x$, $y = 0$ **when** $x = \frac{\pi}{3}$

1. $y = \cos x - 2 \cos^2 x$ 2. $y = \sin x - 2 \sin^2 x$
3. $y = \cos x - 2 \cos^2 x$ 4. $y = \cos x - 2 \cos^3 x$

35. The equation of a curve passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point.

1. $x = y + 1 + e^{-y}$ 2. $y = x + 1 + e^{-x}$
3. $x = y + 1 + e^{-x}$ 4. $y = -(x+1) + e^x$

36. Solve $(1 + y + x^2 y) dx + (x + x^3) dy = 0$

- 1) $xy = -\tan^{-1} x + c$ 2) $x^2 y = -\tan^{-1} x + c$
3) $xy^2 = \tan^{-1} x + c$ 4) $xy^2 = \cot^{-1} x + c$

BERNOULLI EQUATION

37. The solution of $\frac{dx}{dy} + \frac{x}{y} = x^2$ is

- 1) $\frac{1}{y} = cx - x \log x$ 2) $\frac{1}{x} = cy - y \log y$
 3) $\frac{1}{x} = cx - x \log y$ 4) $\frac{1}{y} = cx - y \log x$

38. The solution of

$$\frac{dy}{dx} = 1 - x(y-x) - x^3(y-x)^3 \text{ is}$$

- 1) $(y-x)^2(x^2+1+cx^2)=1$
 2) $(y-x)^2(x^2+1+ce^{x^2})=1$
 3) $(y-x)^2(x^2-1+cx^2)=1$
 4) $(y-x)^2(-x^2-1+ce^{x^2})=1$

39. Solve $x \frac{dy}{dx} + y = x^3 y^6$

- 1) $y^{-5}x^{-5} = \frac{5}{2}x^{-2} + c$ 2) $y^5x^5 = \frac{5}{2}x^2 + c$
 3) $y^{-5}x^3 = \frac{5}{2}x^2 + c$ 4) $y^5x^5 = \frac{5}{2}x + c$

LEVEL-II (H.W.) - KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) 1 | 2) 4 | 3) 1 | 4) 1 | 5) 2 | 6) 1 | 7) 2 |
| 8) 1 | 9) 2 | 10) 2 | 11) 3 | 12) 1 | 13) 3 | 14) 3 |
| 15) 3 | 16) 1 | 17) 2 | 18) 1 | 19) 4 | 20) 1 | 21) 1 |
| 22) 2 | 23) 1 | 24) 2 | 25) 1 | 26) 1 | 27) 1 | 28) 1 |
| 29) 2 | 30) 1 | 31) 1 | 32) 3 | 33) 2 | 34) 1 | 35) 4 |
| 36) 1 | 37) 2 | 38) 4 | 39) 1 | | | |

LEVEL-II (H.W.) - HINTS

- given circle is $(x-a)^2 + (y-a)^2 = a^2$
differentiating w.r.t. 'x'
- $y^2 = 4a(x+a)$ and eliminate a
- $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
find y_1 and y_2 eliminate a, b

4. find y_1 and y_2 eliminate a, b

5. $m = \frac{dy}{dx}$ slope of tangent of hyperbola

$$m_2 = -\frac{c^2}{x^2}; \tan 45^\circ = \frac{m + \frac{c^2}{x^2}}{1 - \frac{mc^2}{x^2}} \Rightarrow \frac{dy}{dx} = \frac{x^2 - c^2}{x^2 + c^2}$$

6. Find y_1 and $\frac{y}{y_1}$

7. Differentiate one time with respect to x and find

$$\frac{x^3}{3x^2} = \frac{c(y+c)^2}{2(y+c)y_1}$$

8. find y_1 and y_2 eliminate a, b

9. find y_1, y_2, y_3 and y_4 eliminate a, b, c, d

10. find y_1 and y_2 eliminate a, b

$$11. dy = \left(x + \frac{1}{x^2}\right) dx$$

$$12. \frac{d^2y}{dx^2} = \log x \Rightarrow \frac{dy}{dx} = \int \log x dx$$

$$13. \int 2x\sqrt{1+x^2} dx + \int 2y\sqrt{1+y^2} dy = c$$

14. divide with xy

$$x^2 dx + y dx + x dy + y^2 dy = 0$$

$$15. x^2 dx + d(xy) + y^2 dy = 0$$

$$16. \int e^{-by} dy = \int e^{ax} dx$$

17. put $y-x = V$ apply variable

$$18. e^{-y} dy = e^{3x} dx$$

$$19. x \sec^2 x dx = y \sec^2 y dy$$

$$20. \sqrt{(1+x^2)(1+y^2)} + xy \frac{dy}{dx} = 0 \text{ separate the terms}$$

$$21. \frac{dy}{dx} = x(y+2) - 3(y+2) = (x-3)(y+2)$$

22. separate the terms

23. $\frac{dx}{x} = \frac{ydy}{1+y^2}$ integrate it
24. $xe^{x^2} \cdot e^y dx = Y dy$
25. $\frac{dy}{dx} = v + \frac{xdv}{dx}$
26. put $\frac{y}{x} = v$
27. $\frac{dy}{dx} - \frac{2y}{x} = 2x^3 + x$, I.F = $e^{\int \frac{-2}{x} dx}$
28. $\frac{dx}{dy} = \frac{x+2y^3}{y^2}$, I.F = $e^{\int \frac{-1}{y^2} dy}$
29. convert the problem to $\frac{dx}{dy} + P(y)x = y$
30. G.E = $\frac{dx}{dy} = \frac{x+1}{2e^{-y}+1}$; $\frac{dx}{dy} - \frac{x}{2e^{-y}+1} = \frac{1}{2e^{-y}+1}$
 I.F = $e^{\int p dy} = e^{\int \frac{-1}{2e^{-y}+1} dy} = e^{\int \frac{-e^y}{2+e^y} dy}$
 $= e^{-\log(2+e^y)} = \frac{1}{2+e^y}$
 G.S = $x \frac{1}{2+e^y} = \int \frac{1}{2+e^y} \cdot \frac{1}{2e^{-y}+1} dy$
 $= \int \frac{e^y dy}{(2+e^y)^2} = \frac{-1}{2+e^y} + c$
31. I.F = e^{ax} , $ye^{ax} = \int c^{(a+m)x} dx = e$
32. dividing with x^3
33. $I.F = e^{\int -\tan x dx}$, solution is
 $y(I.F) = \int (I.F) e^x \sec x dx + c$
34. I.F $e^{\int 2 \tan x} = e^{2 \log(\sec x)} = \sec^2 x$
 $y \sec^2 x = \int \sin x \sec^2 x dx = \int \tan x \sec x dx$
 $= y \sec^2 x = \sec x + c$ at $x = 60^\circ$, $y = 0$
 $0 = 2 + c \Rightarrow c = -2$
 $\Rightarrow y \sec^2 x = \sec x - 2 \Rightarrow y = \cos x - 2 \cos^2 x$
35. $\frac{dy}{dx} = x + y$; $\frac{dy}{dx} - y = x$; I.F = $e^{\int -1 dx} = e^{-x}$

- G.S = $ye^{-x} = \int xe^{-x} dx$
 $ye^{-x} = e^{-x}(-x-1) + c = -(x+1)e^{-x} + c$
 since it passes through the origin $c = 1$
 $\therefore y = -(x+1) + e^x$
36. $1 + y(1+x^2) + x(1+x^2) \frac{dy}{dx} = 0$
 this is linear differential equation then solve
37. $\frac{dx}{dy} + \frac{x}{y} = x^2 \Rightarrow x^{-2} \frac{dx}{dy} + \frac{x^{-1}}{y} = 1$
 $x^{-1} = t \Rightarrow x^{-2} \frac{dx}{dy} = \frac{-dt}{dy}$; $\frac{dt}{dy} - \frac{1}{y} t = -1$
 I.F = $e^{-\int \frac{1}{y} dy} = e^{-\log y} = \frac{1}{y}$
 $t \left(\frac{1}{y} \right) = \int (-1) \frac{1}{y} dy = \log y + c$
 $\Rightarrow x^{-1} = cy - y \log y$
38. put $y - x = z \Rightarrow \frac{dy}{dx} - 1 = \frac{dz}{dx}$
 G.E becomes $\frac{dz}{dx} + zx = -x^3 z^3$
 $\Rightarrow \frac{1}{z^3} \frac{dz}{dx} + \frac{x}{z^2} = -x^3$
 put $z^{-2} = v \Rightarrow -2z^{-3} \frac{dz}{dx} = \frac{dv}{dx} \Rightarrow \frac{dv}{dx} - 2xv = 2x^3$
 I.F = $e^{\int -2x dx} = e^{-x^2}$;
 G.S = $ve^{-x^2} = \int 2x^3 e^{-x^2} dx$; put $-x^2 = t$
 $= \int te^t dt = te^t - e^t + c$; $ve^{-x^2} = (-x^2 - 1)e^{-x^2} + c$
 $\Rightarrow v = -x^2 - 1 + ce^{x^2} \Rightarrow 1 = (y-x)^2 (-x^2 - 1 + ce^{x^2})$
39. $\frac{1}{y^6} \frac{dy}{dx} + \frac{1}{xy^5} = x^2$ Put $y^{-5} = v$
 $\Rightarrow y^{-6} \frac{dy}{dx} = \frac{-1}{5} \frac{dv}{dx}$ $\Rightarrow \frac{dv}{dx} - \frac{5}{x} v = -5x^2$
 I.F = $e^{\int \frac{-5}{x} dx} = \frac{1}{x^5}$; G.S = $y^{-5} x^5 = \frac{5}{2} x^{-2} + c$

 LEVEL - III

VARIABLE SEPARABLE

1. The population $p(t)$ at a time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5p(t) - 450$. if $p(0) = 850$, then the time at which the population becomes zero is

1. $2 \ln 18$ 2. $\ln 9$ 3. $\frac{1}{2} \ln 18$ 4. $\ln 18$

2. Solution of $\frac{dy}{dx} = \frac{x(2 \log x + 1)}{\sin y + y \cos y}$

1) $x \sin x = y^2 \log y + c$ 2) $y \sin y = x^2 \log x + c$
 3) $y \cos y = x \log x + c$ 4) $y \sin y = 2x \log x + c$

3. Solution of $\left(\frac{x+y-a}{x+y-b}\right)\left(\frac{dy}{dx}\right) = \left(\frac{x+y+a}{x+y+b}\right)$

1) $\log[(x+y)^2 - ab] = \frac{2}{b-a}[x-y] + k$

2) $\log[(x+y)^2 + ab] = \frac{1}{b-a}[x+y] + k$

3) $\left(\frac{b-a}{2}\right)\left[\log((x+y)^2 - ab)\right] = x + c$

4) $2 \log(x+y) = \frac{x+y}{b-a} + k$

4. The solution of $\frac{dy}{dx} = \frac{ax+h}{by+k}$ represents a parabola when

1) $a = 0, b = 0$ 2) $a = 1, b = 2$
 3) $a = 0, b \neq 0$ 4) $a = 2, b = 1$

5. The curve for which the normal at any point (x, y) and the line joining origin to that point form an isosceles triangle with x axis as base is

1) an ellipse
 2) a rectangular hyperbola
 3) a circle
 4) parabola

6. The curve in which the slope of tangent at any point equal to the ratio of the abscissa to the ordinate of the point is

1) an ellipse 2) a parabola
 3) a rectangular hyperbola 4) a circle.

7. The solution of the diff. equation

$\frac{dy}{dx} = \frac{3x^2y^4 + 2xy}{x^2 - 2x^3y^3}$ is

1) $\frac{y^2}{x} - x^3y^2 = c$ 2) $\frac{x^2}{y} + x^3y^2 = c$

3) $\frac{x^2}{y^2} + x^3y^3 = c$ 4) $\frac{x^2}{3y} - 2x^3y^2 = c$

8. If f and g are twice differentiable on $(0, 2)$ satisfying and $f(2) = 3, g(2) = 9$, then $f(x) - g(x)$ at $x = 3$ is

1) 8 2) 9 3) 12 4) -8

9. Solution for $\frac{x+y \frac{dy}{dx}}{y-x \frac{dy}{dx}} = x^2 + 2y^2 + \frac{y^4}{x^2}$ is

1) $\frac{y}{x} - \frac{1}{x^2 + y^2} = c$ 2) $\frac{y}{x} + \frac{1}{x^2 + y^2} = c$

3) $\frac{x}{y} + \frac{1}{x^2 + y^2} = c$ 4) $\frac{x}{y} - \frac{1}{x^2 + y^2} = c$

10. If the solution of the differential equation

$\frac{1}{\sin^{-1} x} \left(\frac{dy}{dx}\right) = 1$ is $y = x \sin^{-1} x + f(x) + c$ then $f(x)$ is

1) $1 - x^2$ 2) $\sqrt{1 - x^2}$ 3) $1 + x^2$ 4) $\sqrt{1 + x^2}$

11. The solution of the differential equation

$\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2$ when $y'(0) = 1$ and $y(0) = 0$ is $y =$

1) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$

2) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x$

3) $\frac{-\cos 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x + 1$

4) $\frac{\cos 3x}{9} + \frac{x^4}{12} + \frac{x}{3} - e^x + 1$

12. The acceleration after t sec. of a particle which starts from rest and moves in a straight line is

$$\left[5 - \frac{t}{4}\right] \text{ Cm/s}^2. \text{ The velocity of the particle at}$$

the instant when the acceleration is zero is

- 1) 20 Cm/s 2) 40 Cm/s 3) 50 Cm/s 4) 60 Cm/s

13. The differential equation $\frac{dx}{dy} = \frac{3y}{2x}$ represents a family of hyperbolas (except when it represents a pair of lines) with eccentricity

- 1) $\sqrt{\frac{3}{5}}$ 2) $\sqrt{\frac{5}{3}}$ 3) $\sqrt{\frac{5}{4}}$ 4) $\sqrt{5}$

14. The solution of the equation

$$x \int_0^x y(t) dt = (x+1) \int_0^x ty(t) dt, x > 0 \text{ is}$$

1) $y = \frac{c}{x^3} e^{x^3}$ 2) $y = cx^3 e^{x^3}$

3) $\frac{c}{x^3} e^{-x}$ 4) $y = \frac{c}{x^3} e^{\frac{-1}{x}}$

15. If normal is drawn at a point $P(x, y)$ of a curve meets the X- axis and the Y- axis in points A

and B respectively such that $\frac{1}{OA} + \frac{1}{OB} = 1$,

(where 'O' is the origin). The equation of the curve

1) $(x-1)^2 = cy$ 2) $(x-1)^2 = cx$

3) $(x-1)^2 + (y-1)^2 = c$ 4) $x^2 + y^2 = cx$

16. The slope of the tangent of a curve $y=f(x)$ at $(x, f(x))$ is $2x+1$. The curve passes through the point $(1, 2)$. The area bounded by the

curve, the X-axis and the line $x=1$ insq.units is

- 1) 5/6 2) 6/5 3) 1/6 4) 6

17. The value of $\lim_{x \rightarrow \infty} y(x)$ obtained from the

differential equation $\frac{dy}{dx} = y - y^2$ where

$y(0) = 2$ is

- 1) 0 2) 1 3) ∞ 4) 3

18. If the population of a country doubles in 50 years, the number of years it triples under the assumption that the rate of increase is proportional to the number of inhabitants

- 1) 50 years 2) 60 years 3) 80 years 4) 90 years

19. General solution of

$$(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0 \text{ is}$$

1) $y = c \log |\sin x + \cos x|$

2) $y = c - \log |\sin x + \cos x|$

3) $y - \log |\sin x + \cos x| = c$

4) $y = c - \log |\sin x - \cos x|$

20. The equation of the family of curves which intersect the hyperbola $xy=2$ orthogonally is

1) $y = \frac{x^3}{6} + c$ 2) $y = \frac{x^2}{4} + c$

3) $y = \frac{x^2}{6} - c$ 4) $y = \frac{x^2}{4} - c$

HOMOGENEOUS EQUATIONS

21. The solution of the D. E

$$yy_1 = x \left[\frac{y^2}{x^2} + \frac{f(y^2/x^2)}{f'(y^2/x^2)} \right] \text{ is}$$

1) $f\left(\frac{y^2}{x^2}\right) = cx^2$ 2) $x^2 f\left(\frac{y^2}{x^2}\right) = c^2 y^2$

3) $x^2 f(y^2/x^2) = c$ 4) $f\left(\frac{y^2}{x^2}\right) = cy/x$

22. The curve satisfying $y' = \frac{x^2 + y^2}{x^2 - y^2}, y(1) = 0$

has the slope at the point(1,0) of the curve equal to

1) $-\frac{5}{3}$ 2) -1 3) 1 4) $\frac{5}{3}$

23. A curve passing through the point $(1, 1)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x-axis. The equation of the curve is

1) $x^2 + y^2 + 2x = 0$ 2) $x^2 + y^2 - 2x = 0$

3) $x^2 + y^2 - 2y = 0$ 4) $x^2 + y^2 + 2y = 0$

INTEGRATING FACTORS OF DIFFERENTIAL EQUATION

24. If integrating factor of

$x(1-x^2)dy + (2x^2y - y - ax^3)dx = 0$ is $e^{\int p dx}$ then p is

- 1) $\frac{2x^2+1}{x(1-x^2)}$ 2) $2x^3-1$
 3) $\frac{2x^2-1}{ax^3}$ 4) $\frac{2x^2-1}{x(1-x^2)}$

LINEAR EQUATIONS

25. The tangent at a point 'P' of a curve meets the axis of 'y' in N, the parallel through 'P' to the axis of 'y' meets the axis of X at M, 'O' is the origin. of the area of $\triangle MON$ is constant, then the curve is

- 1) circle 2) parabola
 3) ellipse 4) hyperbola

26. The curve satisfy the differential equation

$$(1-x^2)y' + xy = ax \text{ are}$$

- 1) ellipse and hyperbola
 2) ellipse and parabola
 3) ellipse and straight line
 4) circle and ellipse

BERNOULLI EQUATION

27. The solution of

$$2y \cos y^2 \frac{dy}{dx} - \frac{2}{x+1} \sin y^2 = (x+1)^3 \text{ is}$$

- 1) $\sin y^2 = (x+1)^2 \left[(x+1)^2 + c \right]$
 2) $\sin y^2 = (x+1)^2 \left[\frac{(x+1)^2}{2} + c \right]$
 3) $\sin y^2 = (x+1)^2 \left[\frac{(x+1)^3}{3} + c \right]$
 4) $\sin y^2 = (x+1)^2 \left[\frac{(x+1)^4}{4} + c \right]$

28. A solution of the differential equation

$$\frac{dy}{dx} = \frac{1}{xy(x^2 \sin y^2 + 1)} \quad (C \text{ is arbitrary constant})$$

- 1) $x^2 (\cos y^2 - \sin y^2 - 2ce^{-y^2}) = 2$
 2) $y^2 (\cos x^2 - \sin y^2 + 2ce^{-y^2}) = 2$
 3) $y^2 (\cos y^2 - \sin y^2 - e^{-y^2}) = 4$
 4) $y^2 (\cos y^2 - \sin x^2 + e^{y^2}) = 2$

29. Solve $\left(xy^2 - e^{\frac{1}{x^3}} \right) dx - x^2 y dy = 0$

- 1) $3x^2 = 2y^2 e^{\frac{1}{x^2}} + cx^2$ 2) $3y^2 = 2x^2 e^{\frac{1}{x^3}} + cx^2$
 3) $y = 2x^2 e^{\frac{1}{x^3}} + cx^2$ 4) $x^2 = 2y e^{\frac{1}{x^2}} + cx^2$

30. Solve $\frac{dy}{dx} = x^3 y^3 - xy$

- 1) $(y^2 + 1 + ce^{y^2}) = 1$ 2) $(x^2 + 1 + ce^{x^2}) y^2 = 1$
 3) $(y^2 + 1 + ce^{y^2}) = x$ 4) $(x + 1 + ce^{y^2}) x^2 = 1$

31. General solution of differential equation

$$\frac{dy}{dx} + yg'(x) = g(x)g'(x) \quad \text{where } g(x) \text{ is a function in } x.$$

- 1) $g(x) + \log(1+y+g(x)) = c$
 2) $g(x) + \log(1+y-g(x)) = c$
 3) $g(x) - \log(1+y-g(x)) = c$
 4) $g(x) + \log(y+g(x)) = c$

32. The function $f(\theta) = \frac{d}{d\theta} \int_0^\theta \frac{dx}{1 - \cos \theta \cos x}$ satisfies the differential equation .

- 1) $\frac{df}{d\theta} + 2f(\theta) \cot \theta = 0$ 2) $\frac{df}{d\theta} - 2f(\theta) \cot \theta = 0$
 3) $\frac{df}{d\theta} + 2f(\theta) = 0$ 4) $\frac{df}{d\theta} - 2f(\theta) = 0$

33. If $f(x)$ and $g(x)$ are two solutions of the

differential equation $a \frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} + y = e^x$,

then $f(x) - g(x)$ is the solution of

1) $a^2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = e^x$ 2) $a^2 \frac{d^2 y}{dx^2} + y = e^x$

3) $a \frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} + y = 0$ 4) $a \frac{d^2 y}{dx^2} + y = e^x$

34. The solution of the D. E $\frac{dy}{dx} = \frac{y \cdot f^1(x) - y^2}{f(x)}$

equal to

1) $xy = f(x) + c$ 2) $xy = f(x) + cx$

3) $y(x + c) = f(x)$ 4) $y = f(x) + x + c$

35. The degree of the differential equation satisfying

$\sqrt{1+x^2} + \sqrt{1+y^2} = K(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is

1) 4 2) 3 3) 1 4) 2

36. The solution of $\frac{xdx + ydy}{ydx - xdy} = \frac{x \sin(x^2 + y^2)}{y^3}$ is

1) $\log |\tan(x^2 + y^2)| = \frac{x^2}{y^2} + c$

2) $\log \left| \tan \left(\frac{x^2 + y^2}{2} \right) \right| = \frac{x^2}{y^2} + c$

3) $\log |\tan(x^2 + y^2)| = \frac{y^2}{x^2} + c$

4) $\log \left| \tan \left(\frac{x^2 + y^2}{2} \right) \right| = \frac{y^2}{x^2} + c$

37. The equation of the curve, passing through (2, 5) and having the area of triangle formed by the X-axis, the ordinate of a point on the curve and the tangent at the point is 5 sq. units is

1) $xy = 10$ 2) $x^2 = 10y$
3) $y^2 = 10x$ 4) $xy^{1/2} = 10$

38. The equation of the curve not passing through origin and having portion of the tangent included between the coordinate axes is bisected the point of contact is

1. A parabola
2. An ellipse or a straight line
3. A circle or an ellipse
4. A hyperbola

39. The solution of

$\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$ is

1. $y \cos x + x \cos y + xy = k$
2. $y \sin x + x \sin y + xy = k$
3. $\sin x + \sin y + xy = k$
4. $xy(\sin x + \sin y + 1) = k$

40. The orthogonal trajectories of the family of curves $a^{n-1}y = x^n$ are give by

1. $x^n + n^2 y = \text{constant}$
2. $ny^2 + x^2 = \text{constant}$
3. $n^2 x + y^n = \text{constant}$
4. $n^2 x - y^n = \text{constant}$

LEVEL - III - KEY

1) 1	2) 2	3) 1	4) 3	5) 2	6) 3	7) 2
8) 4	9) 1	10) 2	11) 1	12) 3	13) 2	14) 4
15) 3	16) 1	17) 2	18) 3	19) 2	20) 1	21) 1
22) 3	23) 2	24) 4	25) 4	26) 1	27) 2	28) 1
29) 2	30) 2	31) 2	32) 1	33) 3	34) 3	35) 3
36) 2	37) 1	38) 2	39) 2	40) 2		

LEVEL - III - HINTS

1. $\frac{dp(t)}{dt} = 0.5p(t) - 450; 2 \int \frac{d(p(t))}{p(t) - 900} = \int dt$

$2 \ln |p(t) - 900| = t + c$

$t = 0$ then $2 \ln 50 = 0 + c$ then $c = 2 \ln 50$

$2 \ln |p(t) - 900| = t + 2 \ln 50$

$p(t) = 0 \Rightarrow 2 \ln 900 = t + 2 \ln 50$

$t = 2(\ln 900 - \ln 50) = 2 \ln \left(\frac{900}{50} \right) = 2 \ln 18$

2. apply variable separable

3. put $x + y = z$

4. $\frac{by^2}{2} - \frac{ax^2}{2} + ky - hx - c = 0$ for $a = 0, b \neq 0$

5. conceptual
6. $\frac{dy}{dx} = x/y$
7. $x^2 dy - 2x^3 y^3 dy = 3x^2 y^4 dx + 2xy \cdot dx$
 $2xy dx - x^2 dy + 3x^2 y^4 dx + 2x^3 y^3 dy = 0$
 $\frac{y \cdot 2x - x^2 dy}{y^2} + 3x^2 y^2 dx + 2x^3 y = 0$
 $\Rightarrow d\left(\frac{x^2}{y}\right) + d(x^3 y^2) = 0 \Rightarrow \frac{x^2}{y} + x^3 y^2 = 0$
8. $f''(x) = g''(x)$ on integrating
 $f'(x) = g'(x) + c$ for $x = 1$ $c = -2$
 $\therefore f'(x) = g'(x) - 2$ on integration
 $f(x) = g(x) - 2x + l$ for $x = 2$, $l = -2$
 $\therefore f(x) = g(x) - 2x - 2$; $f(3) - g(3) = -8$
9. Given equation can be written as
 $\frac{xdx + ydy}{(x^2 + y^2)^2} = \frac{ydx - xdy}{y^2} \cdot \frac{y^2}{x^2}$
 $\int \frac{d(x^2 + y^2)}{(x^2 + y^2)^2} = \int \frac{1}{x^2} \cdot d\left(\frac{x}{y}\right) = \frac{2y}{x} - \frac{1}{x^2 + y^2} = c$
10. $\frac{1}{\sin^{-1} x} \frac{dy}{dx} = 1 \Rightarrow dy = \sin^{-1} x dx$ on integration
 $y = x \sin^{-1} x + \sqrt{1 - x^2} \Rightarrow f(x) = \sqrt{1 - x^2}$
11. $y_2 = \sin 3x + e^x + x^2$
 $y_1 = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + c$
 $y_1(0) = \frac{-1}{3} + 1 + c = 1 \Rightarrow c = \frac{1}{3}$
 $y_1 = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + \frac{1}{3}$
 $y = -\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{x}{3} - 1$

12. $\frac{dv}{dt} = 5 - \frac{t}{4} \Rightarrow v = 5t - \frac{t^2}{8} + c$
 since particle was at rest initially $c = 0$, $v = 0$,
 $t = 0$ hence, $v = 5t - \frac{t^2}{8} \therefore$ at 20 seconds.
 $v = 50$ cm per seconds.
13. On integration we get $x^2 - \frac{3y^2}{2} = c$
 If $c \neq 0$ it represents family of hyperbolas with
 $e = \sqrt{\frac{5}{3}}$ ($c > 0$), $e = \sqrt{\frac{5}{2}}$ ($c < 0$)
 If $c = 0$ eqn becomes straight line.
14. $x \int_0^x y(t) dt = (x+1) \int_0^x ty(t) dt$ Diff w.r.t x
 $xy(x) + \int_0^x y(t) dt = (x+1)xy(x) + 1 \int_0^x ty(t) dt$
 again differentiating we get $\frac{1-3x}{x^2} dx = \frac{dy}{y}$
 $\Rightarrow y = \frac{c}{x^3} e^{-\frac{1}{x}}$
15. Equation of normal at (x, y) is
 $(Y - y) = \frac{-1}{\frac{dy}{dx}} (X - x)$
 $OA = x + ym$ $OB = \frac{x + ym}{m} \cdot \frac{1}{OA} + \frac{1}{OB} = 1$;
 $(y-1)\left(\frac{dy}{dx}\right) = -(x-1)$ then integrate
16. $\frac{dy}{dx} = 2x + 1 \Rightarrow y = x^2 + x + c$
 passes thro $(1, 2)$ $c = 0$ area bounded by the curve,
 the X-axis and the line $x = 1$
 $\int_0^1 (x^2 + x) dx = \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_0^1 = \frac{5}{6}$ sq. units
17. $\frac{dy}{dx} = -y(y-1) \Rightarrow \frac{dy}{y(y-1)} = -dx$ and
 $\log\left(\frac{y-1}{y}\right) = -x + c$, $y(0) = 2$
 $\Rightarrow c = -\log 2$; $y = \left(\frac{2}{2 - e^{-x}}\right) \therefore \lim_{x \rightarrow \infty} y(x) = 1$

18. 'x' be the population at a time 't' in years $\frac{dx}{dt}$

$$\frac{dx}{dt} \propto x \Rightarrow \frac{dx}{dt} = kx \text{ integration gives } x = x_0 e^{kt}$$

x_0 is the population at time $t=0$. Since it doubles in fifty years at $t=50$. we must have $x = 2x_0$

$$\text{Hence, } 2x_0 = x_0 e^{50k} \Rightarrow k = \frac{1}{50} \log 2, \text{ to find } t, \text{ it}$$

tripplies $x = 3x_0$ gives $t = \frac{50 \log 3}{\log 2} = 80$ years approximately

19. $\frac{dy}{dx} = \frac{\sin x - \cos x}{\sin x + \cos x} \Rightarrow dy = \tan\left(\frac{\pi}{4} - x\right) dx$

20. Let the family of curves be $y=f(x)$, given $y = \frac{2}{x}$

$$m_1 = \frac{-2}{x^2}, m_2 = \frac{x^2}{2}; \frac{dy}{dx} = \frac{x^2}{2} \Rightarrow dy = \frac{x^2}{2} dx$$

21. put $\frac{y}{x} = V$

22. y' at $(1,0) = 1$

23. Equation of normal at (x_1, y_1) is

$$y - y_1 = \frac{-1}{m}(x - x_1) \text{ from given condition}$$

$$\left| \frac{-my_1 - x_1}{\sqrt{1+m^2}} \right| = y_1 \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Solve the Homogeneous equation and apply initial condition.

24. $\frac{dy}{dx} + \frac{(2x^2 - 1)}{x(1 - x^2)} y = \frac{ax^3}{x(1 - x^2)}$

$$\therefore \frac{dy}{dx} + Py = Q \Rightarrow P = \frac{2x^2 - 1}{x(1 - x^2)}$$

25. Tangent is $Y - y = \frac{dy}{dx}(X - x)$

it meets y-axis in $N \Rightarrow Y = y - x \frac{dy}{dx}$ (putting $X=0$)

$$\text{also } \Delta^{le} MON = \frac{1}{2} OM \cdot ON = \text{constant}$$

$$\therefore \frac{1}{2} x \left(y - x \frac{dy}{dx} \right) = \text{constant}$$

$$\Rightarrow \frac{dy}{dx} - \frac{1}{x} y = \frac{c}{x^2} \Rightarrow \frac{y}{x} = \frac{-c}{2x^2} + k$$

$$\Rightarrow 2kx^2 - 2xy + c = 0$$

it Rep a hyperbola $\therefore \Delta \neq 0, h^2 - ab > 0$

26. The given equation is linear in y can be written as

$$\frac{dy}{dx} + \frac{x}{1 - x^2} y = \frac{ax}{1 - x^2}$$

I.F =

$$e^{\int \frac{x}{1-x^2} dx} = e^{-\frac{1}{2} \log(1-x^2)} = \frac{1}{\sqrt{1-x^2}} \text{ if } -1 < x < 1$$

$$\text{I.F} = \frac{1}{\sqrt{x^2 - 1}} \text{ if } x^2 > 1$$

$$\Rightarrow y \left(\frac{1}{\sqrt{1-x^2}} \right) = \int \frac{ax}{(1-x^2)^{3/2}} dx = \frac{a}{\sqrt{1-x^2}} + c$$

$$\Rightarrow y = a + c\sqrt{1-x^2} \Rightarrow (y-a)^2 + c^2 x^2 = c^2$$

ellipse if $-1 < x < 1$

If $x^2 > 1$ above solution becomes

$$-(y-a)^2 - c^2 x^2 = c^2 \text{ represents hyperbola}$$

27. Let $\sin y^2 = t \Rightarrow 2y \cos y^2 \frac{dy}{dx} = \frac{dt}{dx}$

$$\therefore \frac{dt}{dx} - \frac{2t}{(x+1)} = (x+1)^3$$

$$\text{Solve the linear equation. I.F.} = \frac{1}{(x+1)^2}$$

28. GE $\frac{dx}{dy} = xy(x^2 \sin y^2 + 1)$

$$\Rightarrow \frac{1}{x^3} \frac{dx}{dy} - \frac{1}{x^2} y = y \sin y^2$$

$$\text{put } -\frac{1}{x^2} = u \Rightarrow \frac{du}{dy} + 2uy = 2y \sin y^2$$

$$\text{I.F} = e^{y^2}; \text{ G.S} = ue^{y^2} = \int 2y \sin y^2 e^{y^2} dy + c$$

$$\text{put } y^2 = t \Rightarrow \int e^t \sin t dt + c$$

$$ue^{y^2} = \frac{1}{2} e^{y^2} (\sin y^2 - \cos y^2) + c$$

$$\Rightarrow 2 = x^2 (-\sin y^2 + \cos y^2 - 2ce^{-y^2})$$

29. D.E can be written as $y \frac{dy}{dx} - \frac{y^2}{x} = \frac{-e^{x^3}}{x^2}$

$$\text{put } y^2 = t \Rightarrow y \frac{dy}{dx} = \frac{1}{2} \frac{dt}{dx}$$

$$\text{I.F} = e^{-2 \int \frac{1}{x} dx} = \frac{1}{x^2}$$

$$t \frac{1}{x^2} = \int \frac{1}{x^2} \cdot \frac{-2}{x^2} e^{x^3} dx = \int \frac{-2}{x^4} e^{x^3} dx$$

$$\frac{y^2}{x^2} = \frac{2}{3} e^{x^3} + c \Rightarrow 3y^2 = 2x^2 e^{x^3} + c x^3$$

30. G.E $\frac{dy}{dx} + xy = x^3 y^3 \Rightarrow \frac{1}{y^3} \frac{dy}{dx} + \frac{x}{y^2} = x^3$

$$\text{put } \frac{1}{y^2} = z \Rightarrow \frac{dz}{dx} - 2xz = -2x^3; \quad \text{I.F} = e^{-x^2}$$

$$\text{General solution } ze^{-x^2} = \int -2x^3 e^{-x^2} dx$$

$$\text{put } x^2 = t \Rightarrow ze^{-x^2} = -\int te^{-t} dt \Rightarrow \frac{1}{y^2} e^{-x^2}$$

$$= x^2 e^{-x^2} + e^{-x^2} + c \Rightarrow (x^2 + 1 + ce^{x^2}) y^2 = 1$$

31. put $g(x) - y = v$

32. diff w.r.t. θ

33. $a \frac{d^2 f(x)}{dx^2} + \frac{x^2 df(x)}{dx} + f(x) = e^x$

$$a \frac{d^2 g(x)}{dx^2} + \frac{x^2 dg(x)}{dx} + g(x) = e^x$$

$$\therefore a \frac{d^2}{dx^2} [f(x) - g(x)] + x^2 \frac{d}{dx} [f(x) - g(x)] + [f(x) - g(x)] = 0$$

34. $\frac{dy}{dx} - \frac{yf'(x)}{f(x)} = \frac{-y^2}{f(x)}$

35. Putting $x = \tan \alpha, y = \tan \beta$ and taking transformations $\tan^{-1} x - \tan^{-1} y = 2 \cot^{-1} k$

$$\text{diff. } \frac{1}{1+x^2} - \frac{1}{1+y^2} \frac{dy}{dx} = 0$$

36. $\frac{x dx + y dy}{y dx - x dy} = \frac{x \sin(x^2 + y^2)}{y^3}$

$$\Rightarrow \frac{\frac{1}{2} d(x^2 + y^2)}{y^2 d\left(\frac{x}{y}\right)} = \frac{x \sin(x^2 + y^2)}{y^3}$$

$$\Rightarrow \csc(x^2 + y^2) d(x^2 + y^2) = 2 \left(\frac{x}{y}\right) d\left(\frac{x}{y}\right)$$

$$\text{on integration } \ln \left| \tan \left| \frac{x^2 + y^2}{2} \right| \right| = \frac{x^2}{y^2} + c$$

37. Verify the point (2, 5). It lies on $xy=10$.

39. $\tan y = t$

$$\sec^2 y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + 2xt = x^3$$

$$\text{If } \int 2x dx = e^{x^2}$$

$$\text{Solution } te^{x^2} = \int x^3 e^{x^2} dx$$

$$\tan y = \frac{x^2}{2} - \frac{1}{2} + ce^{-x^2}$$

40. $n_1 a^{n-1} y = x^n; \quad nx^{n-1} y = x^n y_1$

$$ny = x \frac{dy}{dx}; \quad \text{Replace } \frac{dy}{dx} = \frac{-dx}{dy}$$

$$ny dy = -x dx; \quad \text{Solution } x^2 + ny^2 = c$$

LEVEL-IV

VARIABLE SEPARABLE

1. Match the following

List-I

List-II

- a) A straight line with slope 2
b) Parabola whose axis is parallel to y-axis
c) Rectangular hyperbola whose asymptotes are $xy=0$
d) Satisfying the curve $y = e^{cx}$

	a	b	c	d
1.	1	3	4	2
2.	4	3	2	1
3.	1	2	3	4
4.	4	1	2	3

2. Match the following

List-I

List-II

- 1 The solution of D.E $(e^y + 1)\cos x dx + e^y \sin x dy = 0$ which passes through $\left(\frac{\pi}{6}, 0\right)$ is
2. Equation of the curve whose length of subtangent is K is
3. The general solution of $x \log x \frac{dy}{dx} + y = 1$ is
4. The point to which the origin is to be shifted to convert

	1	2	3	4
1.	b	c	a	d
2.	e	a	c	b
3.	a	b	c	d
4.	e	a	d	c

e) $(1 + e^y)\sin x = 1$

3. Match the following

Differential equations Its solutions

- a) $yy_1 = \sec^2 x$
b) $y_1 = x \sec y$
c) $y_1 + (2y \tan x) = \sin x$
d) $xy_1 + y = \cos x$
1. $y \sec^2 x = \sec x + c$
2. $xy = \cos y + c$
3. $xy = \sin x + c$
4. $y^2 = 2 \tan x + c$
5. $x^2 = 2 \sin y + c$

	a	b	c	d
1.	3	2	5	4
2.	4	1	2	3
3.	4	5	1	3
4.	3	5	1	2

4. I. The solution of $(1 + x^2)dy = xydx$ is

$y = c\sqrt{1 + x^2}$

II. The solution of $y \frac{dy}{dx} + x = 1$ is $y^2 = (1 - x)^2$

Which of the above statements is correct

1. only I
2. only II
3. Both I and II
4. Neither I nor II

5. Match the following

List-I

List-II

The D.E's Of

- A) $y = Ae^x + Be^{-x}$ (A, B arbitrary constants)
B) $y = a \sin px + b \cos px$ (a, b arbitrary constants)
C) $y = ax^2 + bx + c$ (a, b, c arbitrary constants)
D) $y = a \sin(mx + b)$ (a, b arbitrary constants)
1. $y_2 + m^2 y = 0$
2. $y_3 = 0$
3. $y_2 - y = 0$
4. $y_2 + p^2 y = 0$
5. $y_2 - m^2 y = 0$

	A	B	C	D
1.	5	2	3	4
2.	2	3	4	5
3.	4	3	2	1
4.	3	4	2	1

6. I. The integrating factor of D.E

$$1 + (x \tan y - \sec y) \frac{dy}{dx} = 0 \text{ is } \sec y$$

II. The I.F of differential equation

$$x(x-1) \frac{dy}{dx} - y = x^2(x-1)^2 \text{ is } \frac{x-1}{x}$$

Which of the above are correct

1. only I 2. only II
3. Both I and II 4. Neither I nor II

7. Assertion(A): The solution of the equation

$$x dx + y dy = \frac{x dy - y dx}{x^2 + y^2} \text{ is}$$

$$2 \tan^{-1} \frac{y}{x} = x^2 + y^2 + c$$

$$\text{Reason(R): } d\left(\tan^{-1} \frac{y}{x}\right) = \frac{x dy - y dx}{x^2 + y^2}$$

1. A and R both are true and R is correct explanation of A
2. A and R both are true but R is not the correct explanation of A
3. Only A is true 4. Only R is true

Integrating Factors of Differential equation

8. Match the following

The Integrating factor of the D.E

List-I

List-II

A) $\frac{dy}{dx} - y \cot x = \cos ecx$ 1. $x \sin x$

B) $x \log x \frac{dy}{dx} + y = 2 \log x$ 2. xe^x

C) $x \sin x \frac{dy}{dx} +$ 3. $\cos ecx$

$$y(x \cos x + \sin x) = \sin x$$

D) $x \frac{dy}{dx} + y(1+x) = 1$ 4. $-\cos ecx$

5. $\log x$

	A	B	C	D
1.	3	5	1	4
2.	3	5	1	2
3.	3	1	5	4
4.	2	3	4	5

LINEAR EQUATIONS

9. I. The solution of D.E $(x+2y^3) \frac{dy}{dx} = y$ is

$$x = y^3 + cy.$$

II. The solution of D.E $\frac{dy}{dx} + y \tan x = \cos^2 x$ is

$$y \sec x = c + \sin x.$$

Which of the above statements is true

1. only I 2. only II
3. Both I and II 4. Neither I nor II

10. Assertion(A): The general solution of

$$\frac{dy}{dx} - \frac{2}{x+1} y = (x+1)^3 \text{ is } \frac{2y}{(x+1)^2} = (x+1)^2 + c$$

Reason(R): general solution of D.E is

$$y(I.F) = \int Q(I.F) dx$$

1. A and R both are true and R is correct explanation of A
2. A and R both are true but R is not the correct explanation of A
3. Only A is true
4. Only R is true

INTEGRATING FACTORS OF DIFFERENTIAL EQUATION

11. Assertion(A): I.F of $\frac{dy}{dx} + y = x^2$ is e^x

Reason(R): $\frac{dy}{dx} + p(x).y = Q(x)$, I.F is

$$\int p(x) dx \text{ Which one of the above is true}$$

1. A true, R false 2. A false, R true
3. A true, R true
4. A false, R false

LINEAR EQUATIONS

12. Observe the following statements:

I: $dy + 2xy dx = 2e^{-x^2} dx \Rightarrow ye^{x^2} = 2x + c$

II: $ye^{x^2} + 2x = c \Rightarrow dx = (2e^{x^2} - 2xy) dy$

which of the following is correct statement?

1. Both I, II are true
2. Neither I nor II
3. I is true II is false
4. I is false II is true

13. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$ where c_1 and c_2 are arbitrary constants is

1. $y^1 = y^2$
2. $y^{11} = y^1 y$
3. $yy^{11} = y^1$
4. $yy^{11} = (y^1)^2$

14. Solution of the differential equation

$$\cos x dy = y(\sin x - y) dx, 0 < x < \frac{\pi}{2} \text{ is}$$

1. $y \sec x = \tan x + c$
2. $y \tan x = \sec x + c$
3. $\tan x = (\sec x + c)y$
4. $\sec x = (\tan x + c)y$

15. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$ then $y(\ln 2)$ is equal to

1. 5
2. 13
3. -2
4. 7

16. The population $P(t)$ at time t of a certain mouse species satisfies the differential equation

$$\frac{dp(t)}{dt} = 0.5(t) - 450. \text{ If } p(0) = 850, \text{ then the}$$

time at which population becomes zero is

1. $2 \log 18$
2. $\log 9$
3. $\frac{1}{2} \log 18$
4. $\log 18$

17. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production P wrt additional number of workers

x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is

- 1) 2500
- 2) 3000
- 3) 3500
- 4) 4000

LEVEL - IV - KEY

- 1) 4
- 2) 2
- 3) 3
- 4) 1
- 5) 4
- 6) 1
- 7) 1
- 8) 2
- 9) 3
- 10) 1
- 11) 3
- 12) 3
- 13) 4
- 14) 4
- 15) 4
- 16) 1
- 17) 3

LEVEL - IV - HINTS

1. a. $m = 2 \Rightarrow \frac{dy}{dx} = 2$
- b. $y = ax^2 + bx + c \Rightarrow y_3 = 0$
- c. $xdy + ydx = 0$
- d. $y \log y = xy,$
2. 1. Variable separable
2. Length of sub tangent $\frac{y}{y_1} = k$
3. LDE
4. Non homogeneous equation
3. A. variable separable
- B. variable separable
- C. LDE D. LDE
4. I. separate the variables
- II. LDE
5. A. Eliminate constant A,B
- B. Eliminate constant a,b
- C. Eliminate constant a,b,c
- D. Eliminate constant a,b

6. I. $\frac{dx}{dy} \cos y = 1 - x \sin y$

$$\frac{dx}{dy} = \frac{1}{\cos y} - \frac{x \sin y}{\cos y} \Rightarrow \frac{dx}{dy} + x \tan y = \sec y$$

$$\text{I.F} = e^{\int \tan y dy} = e^{\int \frac{\sin y}{\cos y} dy} = e^{-\log \cos y} = \sec y$$

II. $\text{I.F} = e^{\log x - \log(x-1)} = e^{\frac{\log x}{x-1}} = \frac{x}{x-1}$

7. A. $\int \frac{x dy - y dx}{(1 + y^2/x^2)^2} = \int x dy + y dx$

8. a. $\text{I.F} = e^{\int \cot x dx} = e^{\int \frac{\cos x}{\sin x} dx} = e^{\log \sin x} = \sin x$

b. $\text{I.F} = e^{\int \frac{1}{x \sin x} dx} = e^{\log(\sin x)} = \log x$

c. $\text{I.F} = e^{\int \frac{x \cos x + \sin x}{x \sin x} dx} = e^{\int \log x \sin x dx} = x \sin x$

d. $\text{I.F} = e^{\int \frac{1+x}{x} dx} = e^{\int \left(\frac{1}{x} + 1\right) dx} = e^{\log x + x} = x e^x$

9. I. Linear differential equation $\frac{dx}{dy} = \frac{x}{y} + 2y^2$

II. LDE is in the form of $\frac{dy}{dx} + py = Q$

10. Reason is correct explanation of A

$$\Rightarrow y(\text{I.F}) = \int Q(\text{I.F}) dx$$

11. A and R both are true

12. I. $\frac{dy}{dx} + 2xy = 2e^{-x^2}$ (LDE)

II. $ye^{x^2} + 2x = c \Rightarrow \frac{d}{dx}(ye^{x^2} + 2x) = \frac{d}{dx}(c)$

$$(ye^{x^2} \cdot 2x + 2) = -ex^2 \frac{dy}{dx}$$

$$(ye^{x^2} \cdot 2x + 2) dx + e^{x^2} dy = 0$$

13. $y = c_1 e^{c_2 x} \dots (1); y^1 = c_2 c_1 e^{c_2 x}; y^1 = c_2 y \dots (2)$

$$y^{11} = c_2 y^1 \text{ from (2) } c_2 = \frac{y^1}{y}$$

$$\text{so } y^{11} = \frac{(y^1)^2}{y} \Rightarrow yy^{11} = (y^1)^2$$

14. $\cos x dy = y(\sin x - y) dx;$

$$\frac{dy}{dx} = y \tan x - y^2 \sec x;$$

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \tan x = -\sec x;$$

$$\text{let } \frac{1}{y} = t$$

$$-\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dy}{dx} - \tan x = -\sec x \Rightarrow \frac{dt}{dx} + (\tan x)t = \sec x$$

$$\text{I.F} = \sec x \text{ solution is } t(\text{I.F}) = \int (I.F) \sec x dx$$

15. $\frac{dy}{dx} = y + 3 \Rightarrow \frac{dy}{y+3} = dx \text{ in } (y+3) = x + c$

$$x = 0 \Rightarrow y = 2 \Rightarrow \ln 5 = 0 + c \quad c = \ln 5$$

$$\ln(y+3) = x + \ln 5$$

$$y+3 = e^{x+\ln 5} \Rightarrow y+3 = e^{\ln 2 + \ln 5}$$

$$y+3 = 10 \Rightarrow y = 7$$

16. $\frac{dp(t)}{dt} = 0.5(t) - 450$

$$\Rightarrow \frac{2dp(t)}{p(t)-900} = \int dt$$

17. $\int dP = \int (100 - 12\sqrt{x}) dx$

